

initial parameter values

Layer	weights	Biases
Layer 1 (2→2)	$W_{1,11} = 0.1$, $W_{1,12} = 0.3$, $W_{1,21} = 0.2$ $W_{1,22} = 0.4$	$b_{1,1} = 0.1$ $b_{1,2} = 0.1$
Layer 2 (2→2)	$W_{2,11} = 0.3$, $W_{2,12} = 0.5$, $W_{2,21} = 0.4$ $W_{2,22} = 0.2$	$b_{2,1} = 0.1$ $b_{2,2} = 0.1$
Layer 3 (2→2)	$W_{3,11} = 0.2$, $W_{3,12} = 0.4$, $W_{3,21} = 0.3$ $W_{3,22} = 0.1$	$b_{3,1} = 0.1$ $b_{3,2} = 0.1$
output (2→1)	$W_{0,1} = 0.5$, $W_{0,2} = 0.3$	$b_0 = 0.1$

forward pass

$z = W \cdot a_{\text{prev}} + b$ followed by activation $a = \sigma(z)$.

Layer 1

$$z_{1,1} = W_{1,11} \cdot x_1 + W_{1,12} \cdot x_2 + b_{1,1}$$

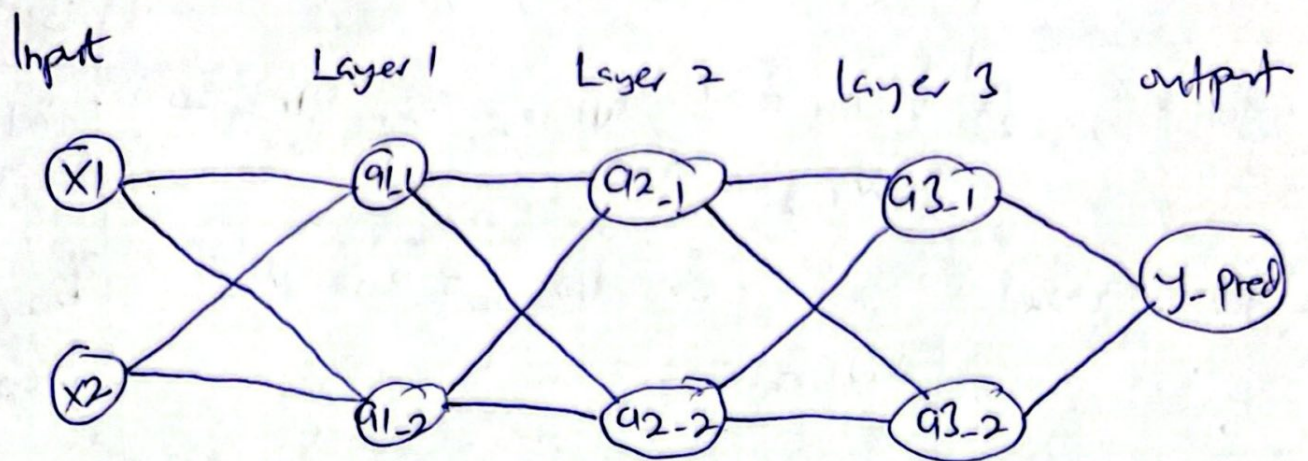
$$= 0.1(0.5) + 0.3(0.8) + 0.1 = 0.05 + 0.24 + 0.1 = \underline{\underline{0.390000}}$$

$$z_{1,2} = W_{1,21} \cdot x_1 + W_{1,22} \cdot x_2 + b_{1,2} = 0.2(0.5) + 0.4(0.8) + 0.1 =$$

$$0.1 + 0.32 + 0.1$$

$$\Rightarrow \underline{\underline{0.520000}}$$

Assignment 1



x_1, x_2 : Inputs y : target output $\hat{y} = a_{out}$: predicted output.

$$\text{Loss} = \frac{1}{2} (y\text{-pred} - y)^2, \quad y = 1.0$$

$z_{i,j}$: pre activation of neuron i in layer j ; $a_{ij} = \sigma(z_{ij})$
Activation.

Sigmoid: $\sigma(z) = \frac{1}{1 + e^{-z}}$ derivative $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Loss, $L = \frac{1}{2} (\hat{y} - y)^2$ $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$

Inputs and target are chosen as $x_1 = 0.5$ $x_2 = 0.8$ $y = 1.0$.

All weights are small positive values between 0.1 and 0.5 and all biases are 0.1

Layer 3

$$\begin{aligned} z_{3,1} &= w_{3,11} \cdot a_{2,1} + w_{3,12} \cdot a_{2,2} + b_{3,1} \\ &= 0.128786 + 0.245580 + 0.1 = \underline{\underline{0.474365}} \end{aligned}$$

$$\begin{aligned} z_{3,2} &= w_{3,21} \cdot a_{2,1} + w_{3,22} \cdot a_{2,2} + b_{3,2} \\ &= 0.3(0.643929) + 0.1(0.613949) + 0.1 \\ &= 0.193179 + 0.061395 + 0.1 \\ &= \underline{\underline{0.354574}} \end{aligned}$$

$$a_{3,1} = \sigma(0.474365) = 0.616416$$

$$a_{3,2} = \sigma(0.354574) = 0.587726$$

Output Layer

$$\begin{aligned} z_{out} &= w_{0,1} \cdot a_{3,1} + w_{0,2} \cdot a_{3,2} + b_0 \\ &= 0.5(0.616416) + 0.3(0.587726) + 0.1 \\ &= 0.308208 + 0.176318 + 0.1 = \underline{\underline{0.584526}} \end{aligned}$$

$$\begin{aligned} \hat{y} &= a_{out} = \sigma(0.584526) = \frac{1}{1 + e^{-0.584526}} = 0.642108 \\ \hat{y} &= 0.642108 \end{aligned}$$

$$a_{1,1} = \sigma(0.390000) = \frac{1}{1 + e^{-0.790000}} = 0.596283$$

$$\cancel{a_{2,2} = \sigma(0.463943) = \frac{1}{1 + e^{-0.463943}} = 0.613949}$$

$$a_{1,2} = \sigma(0.520000) = \frac{1}{1 + e^{-0.520000}} = 0.627148$$

Layer 2

$$z_{2,1} = w_{2,11} \cdot a_{1,1} + w_{2,12} \cdot a_{1,2} + b_{2,1}$$

$$= 0.3(0.596283) + 0.5(0.627148) + 0.1$$

$$= 0.178885 + 0.313574 + 0.1 = \underline{\underline{0.592459}}$$

$$z_{2,2} = w_{2,21} \cdot a_{1,1} + w_{2,22} \cdot a_{1,2} + b_{2,2}$$

$$= 0.4(0.596283) + 0.2(0.627148) + 0.1$$

$$= 0.238513 + 0.125430 + 0.1$$

$$= \underline{\underline{0.463943}}$$

$$a_{2,1} = \sigma(0.592459) = \frac{1}{1 + e^{-0.592459}} = \underline{\underline{0.643929}}$$

$$a_{2,2} = \sigma(0.463943) = \frac{1}{1 + e^{-0.463943}} = \underline{\underline{0.613949}}$$

Loss

$$L = \frac{1}{2} (\hat{y} - y)^2 = \frac{1}{2} (0.642108 - 1)^2 = \frac{1}{2} (-0.357892)^2$$
$$L = \underline{\underline{0.064043}}$$

Back Propagation

The quantity at every layer is the pre activation gradient $\delta = \frac{\partial L}{\partial z}$. Once δ is known for a layer, the gradient of every weight and bias feeding that layer follows. δ is obtained by application of the chain rule.

(i) Pre activation gradient : $\frac{\partial L}{\partial z_{ii}} = \frac{\partial L}{\partial a_{ii}} \cdot \sigma'(z_{ii})$

$$\Rightarrow \frac{\partial L}{\partial a_{ii}} \cdot a_{ii}(1 - a_{ii})$$

(ii) Parameter gradients : $\frac{\partial L}{\partial w_{ii}} = \frac{\partial L}{\partial z_{ii}} \cdot a_{i-1,i}$

$$\frac{\partial L}{\partial b_{ii}} = \frac{\partial L}{\partial z_{ii}}$$

(iii) Passing back one layer.

$$\frac{\partial L}{\partial a_{i-1,i}} = \sum_j \frac{\partial L}{\partial z_{ji}} \cdot w_{ji}$$

Step 1 output Layer

$$\delta L / \delta \hat{y} = \hat{y} - y = 0.642108 - 1.0 = \underline{\underline{-0.357892}}$$

$$\sigma'(z_{out}) = \hat{y} (1 - \hat{y}) = 0.642108 (1 - 0.642108) \\ = \underline{\underline{0.229805}}$$

$$\delta_{out} = \frac{\delta L}{\delta z_{out}} = (-0.357892) (0.229805) = \underline{\underline{-0.082245}}$$

$$\delta L / \delta w_{0,1} = \delta_{out} \cdot a_{3,1} = (-0.082245) (0.616416) \\ = \underline{\underline{-0.050697}}$$

$$\delta L / \delta w_{0,2} = \delta_{out} \cdot a_{3,2} = (-0.082245) (0.587126) \\ = \underline{\underline{-0.048338}}$$

$$\delta L / \delta b_0 = \delta_{out} = \underline{\underline{-0.082245}}$$

Step 2 - Layer 3

Gradient with respect to the layer-3 activation

$$\delta L / \delta a_{3,1} = \delta_{out} \cdot w_{0,1} = (-0.082245) (0.5) = \underline{\underline{-0.041123}}$$

$$\delta L / \delta a_{3,2} = \delta_{out} \cdot w_{0,2} = (-0.082245) (0.3) \\ = \underline{\underline{-0.024674}}$$

pre activation gradient

$$\sigma'(z_{3,1}) = 0.616416 (1 - 0.616416) = \underline{\underline{0.236447}}$$

$$\sigma'(z_{3,2}) = 0.587726 (1 - 0.587726) = \underline{\underline{0.242304}}$$

$$\delta_{3,1} = \delta L / \delta z_{3,1} = (-0.041123)(0.236447) = \underline{\underline{-0.009723}}$$

$$\delta_{3,2} = \delta L / \delta z_{3,2} = (-0.024674)(0.242304) = \underline{\underline{-0.005979}}$$

parameter gradients

$$\begin{aligned} \delta L / \delta w_{3,11} &= \delta_{3,1} \cdot a_{2,1} = (-0.009723)(0.643925) \\ &= \underline{\underline{-0.006261}} \end{aligned}$$

$$\begin{aligned} \delta L / \delta w_{3,12} &= \delta_{3,1} \cdot a_{2,2} = (-0.009723)(0.613945) \\ &= \underline{\underline{-0.005970}} \end{aligned}$$

$$\begin{aligned} \delta L / \delta w_{3,21} &= \delta_{3,2} \cdot a_{2,1} = (-0.005979)(0.643925) \\ &= \underline{\underline{-0.003850}} \end{aligned}$$

$$\delta L / \delta w_{3,22} = \delta_{3,2} \cdot a_{2,2} = (-0.005979)(0.613945)$$

$$\delta L / \delta b_{3,1} = \delta_{3,1} = \underline{\underline{-0.009723}}$$

$$\delta L / \delta b_{3,2} = \delta_{3,2} = \underline{\underline{-0.005979}}$$

Step 3 - Layer 2

Each layer 2 activation feeds both layer 3 neurons
so the contributions are summed.

$$\begin{aligned}\delta L / \delta a_{2,1} &= \delta_{3,1} \cdot w_{3,11} + \delta_{3,2} \cdot w_{3,21} \\ &= (-0.009723)(0.2) + (-0.005979)(0.3) \\ &= -0.001945 - 0.001794 = \underline{\underline{-0.003738}}\end{aligned}$$

$$\begin{aligned}\delta L / \delta a_{2,2} &= \delta_{3,1} \cdot w_{3,12} + \delta_{3,2} \cdot w_{3,22} = \\ &= (-0.009723)(0.4) + (-0.005979)(0.1) \\ &= -0.003889 - 0.000598 = \underline{\underline{-0.004487}}\end{aligned}$$

$$\begin{aligned}\sigma'(z_{2,1}) &= 0.643929 (1 - 0.643929) \\ &= \underline{\underline{0.229284}}\end{aligned}$$

$$\sigma'(z_{2,2}) = 0.613949 (1 - 0.613949) = \underline{\underline{0.237016}}$$

$$\begin{aligned}\delta_{2,1} = \delta L / \delta z_{2,1} &= (-0.003738)(0.229284) \\ &= \underline{\underline{-0.000857}}\end{aligned}$$

$$\delta_{2,2} = \delta L / \delta z_{2,2} = (-0.003738)(0.237016) =$$

$$\delta L / \delta w_{2,1} = \delta_{2,1} \cdot a_{1,1} = (-0.000857)(0.556253) \\ = \underline{\underline{-0.000511}}$$

$$\delta L / \delta w_{2,12} = \delta_{2,1} \cdot a_{1,2} = (-0.000857)(0.627148) \\ = \underline{\underline{-0.000538}}$$

$$\delta L / \delta w_{2,21} = \delta_{2,2} \cdot a_{1,1} = (-0.001064)(0.556253) \\ = \underline{\underline{-0.000634}}$$

$$\delta L / \delta w_{2,22} = \delta_{2,2} \cdot a_{1,2} = (-0.001064)(0.627148) \\ = \underline{\underline{-0.000667}}$$

$$\delta L / \delta b_{2,1} = \delta_{2,1} = -0.000857$$

$$\delta L / \delta b_{2,2} = \underline{\underline{-0.001064}}$$

Step 4 - Layer 1

$$\begin{aligned}\delta L / \delta a_{11} &= \delta_{21} \cdot W_{2,11} + \delta_{22} \cdot W_{2,12} \\ &= (-0.000857)(0.3) + (-0.001064)(0.4) \\ &= -0.000257 - 0.000425 = \underline{\underline{-0.000683}}\end{aligned}$$

$$\begin{aligned}\delta L / \delta a_{12} &= \delta_{21} \cdot W_{2,12} + \delta_{22} \cdot W_{2,22} \\ &= (-0.000857)(0.5) + (-0.001064)(0.2) \\ &= -0.000429 - 0.000213 = \underline{\underline{-0.000642}}\end{aligned}$$

$$\sigma'(z_{11}) = 0.596283(1 - 0.596283) = \underline{\underline{0.240730}}$$

$$\sigma'(z_{12}) = 0.627148(1 - 0.627148) = \underline{\underline{0.233833}}$$

$$\begin{aligned}\delta_{11} &= \delta L / \delta z_{11} = (-0.000683)(0.240730) \\ &= \underline{\underline{-0.000164}}\end{aligned}$$

$$\begin{aligned}\delta_{12} &= \delta L / \delta z_{12} = (-0.000642)(0.233833) \\ &= \underline{\underline{-0.000150}}\end{aligned}$$

Step 4 Layer 1 Continued

The inputs to Layer 1 are raw inputs x .

$$\begin{aligned}\frac{\partial L}{\partial w_{11}} &= \delta_{1,1} \cdot x_1 = (-0.000164)(0.5) \\ &= \underline{\underline{-0.000082}}\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial w_{1,2}} &= \delta_{1,1} \cdot x_2 = (-0.000164)(0.8) \\ &= \underline{\underline{-0.000131}}\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial w_{1,21}} &= \delta_{1,2} \cdot x_1 = (-0.000150)(0.5) \\ &= \underline{\underline{-0.000075}}\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial w_{1,22}} &= \delta_{1,2} \cdot x_2 = (-0.000150)(0.8) \\ &= \underline{\underline{-0.000120}}\end{aligned}$$

$$\frac{\partial L}{\partial b_{11}} = \delta_{11} = \underline{\underline{-0.000164}}$$

$$\frac{\partial L}{\partial b_{12}} = \delta_{1,2} = \underline{\underline{-0.000150}}$$